

Birla Institute of Technology & Science - Pilani, Hyderabad Campus

Second Semester 2022-23

BITS F464 – Machine Learning

Mid Semester Examination

Type: Closed

Time: 90 mins

Max Marks: 60

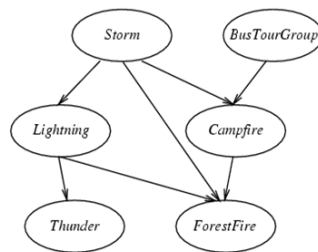
Date: 12.03.2023

All parts of the same question should be answered together.

1. Suppose we train several classifiers to learn $f : X \rightarrow Y$, where X is the feature vector $X = \langle X_1, X_2, X_3, X_4 \rangle$ and Y is a target attribute $\langle 1, 0 \rangle$. Which of the following classifiers contains sufficient information to find out the joint probability distribution of X_1, X_2, X_3, X_4, Y i.e., $P(X_1, X_2, X_3, X_4, Y)$? If you answer yes, give a brief sketch of how. If you answer no, state what is missing. [6 Marks]

a. Fisher's linear discriminant b. Logistic Regression c. Naive Bayes Classifier

2. As seen in the class, consider the Bayes network below, defined over six Boolean variables



a. How many parameters at the least are required to be computed from the training data to define the joint probability distribution of all the attributes i.e., Storm, BusTourGroup, Lightning, Campfire, Thunder, ForestFire. [6 Marks]

Hint: The parameters here should be understood as probability of an attribute taking a values given each parent is taking certain value. The examples of parameters are $P(\text{Campfire} = 1 / \text{Storm} = 0 \text{ and } \text{BusTourGroup} = 1)$, $P(\text{Thunder} = 1 / \text{Lightning} = 1)$ etc. for this Bayes Net. Please note that the parameter $P(\text{Campfire} = 0 / \text{Storm} = 0 \text{ and } \text{BusTourGroup} = 1)$ can be computed once we know the parameter $P(\text{Campfire} = 1 / \text{Storm} = 0 \text{ and } \text{BusTourGroup} = 1)$.

b. Given a huge data set, it is found that the joint distribution of all the attributes Storm, BusTourGroup, Lightning, Campfire, Thunder, ForestFire is a uniform distribution. What would be the prediction of ForestFire (1 or 0) given the test data instance – (Storm = 1, BusTourGroup = 0, Lightning = 1, Campfire = 0, Thunder = 1). [8 Marks]

3. Consider the following vector of binary attributes: (shortbread, lager, whiskey, porridge, football)

A vector $x = (1, 0, 1, 1, 0)^T$ would describe that a person likes shortbread, does not like lager, drinks whiskey, eats porridge, and has not watched England play football. Together with each vector x , there is a label nat describing the nationality of the person, $\text{dom}(\text{nat}) = \{\text{scottish}, \text{English}\}$ [8 Marks]

0	1	1	1	0	0
0	0	1	1	1	0
1	1	0	0	0	0
1	1	0	0	0	1
1	0	1	0	1	0

(a)

1	1	1	1	1	1	1
0	1	1	1	1	0	0
0	0	1	0	0	1	1
1	0	1	1	1	1	0
1	1	0	0	1	0	0

(b)

(a): English tastes for 6 people over attributes: (shortbread; lager; whiskey; porridge; football).

Each column represents the tastes of an individual.

(b): Scottish tastes for 7 people.

Given the characteristics of John, $(1, 0, 1, 1, 0)^T$, classify whether he is an English or Scottish man.

4.a. The following data set consists of a subset of the 3-dimensional 0/1 vectors labeled +1. The remaining 0/1 vectors are labeled -1.

{011, 010, 110, 111}

Note: 011 represents the first feature is 0, second feature is 1 and third feature is 1. And it is labeled +1.

Note: 001 does not belong to {010, 011, 100, 111} and hence 001 is labeled -1.

How confident are you to apply perceptron algorithm to build a classifier? Your answer should not be dependent on applying the algorithm and seeing output of the algorithm. [4 Marks]

4.b. Suppose you are given four training examples $\langle (2, 5, +), (-1, 4, -), (3, -8, +), (4, 7, -) \rangle$. A basis function $\phi((x_1, x_2)^T)$ is applied on these training examples for which the kernel is $K(x, y) = (x_1 y_1 + x_2 y_2)^2$ where $x = (x_1, x_2)^T$ and $y = (y_1, y_2)^T$. The perceptron classifier is built in the transformed space, that is in the space where data points are transformed using the basis function ϕ . And it surprised to figure out that the transformed points are being used exactly 2, 3, 1, 5 times by the perceptron algorithm to find out the discriminant function. (i.e., the first transformed training point is used exactly 2 times, the second transformed training point is used exactly 3 times etc.). Given the testing point $(5, -11)$ in the original feature space, find out whether this point is classified as a positive point (+) or a negative point (-). [8 Marks]

Note: You should not use the explicit basis function, ϕ (though you know) to solve this problem.

4.c. Find the mapping $\phi(x)$ that gives rise to the kernel $K(x, y) = (x_1 y_1 + x_2 y_2)^2$. [6 Marks]

Note: The solution of part b should not be used to solve the part a of this question.

5. a. Given training data set $\{(x_1, t_1), (x_2, t_2), \dots, (x_N, t_N)\}$, derive the optimal parameters, w , using the gradient descent method for the logistic regression model.

$$p(t = 1/x) = \frac{1}{1 + e^{-w^T x}}$$

Note: You should put up detailed mathematical steps that are required in this solution. You may assume there are two features for this problem i.e., $x_n = (x_{n1}, x_{n2})$. You need not prove the convergence of gradient descent algorithm. [8 Marks]

5.b. Given training data set $\{(x_1, t_1), (x_2, t_2), \dots, (x_N, t_N)\}$, where there are only two features i.e., $x_n = (x_{n1}, x_{n2})$. Suppose a logistics regression model has been built

$$p(t = 1/x) = \frac{1}{1 + e^{-w^T x}}$$

where $w = (w_1, w_2)^T$. You are given the decision boundary of the model as $3x_1 + 7x_2 = 5$ that has been obtained by taking the probability threshold as 0.8. You are not given the model itself. Find out the parameters, w_1 and w_2 , of the logistic regression model. [6 Marks]